

Name: _____

Hammack

Score: _____

Directions: Do problem 2, and **one** of problems 1 or 3.

1. Let H be a subgroup of a group G with the property that $ghg^{-1} \in H$ for every $g \in G$ and $h \in H$. Prove that $gH = Hg$ for every $g \in G$. (i.e., show that the left and right cosets of H are identical.)

2. List the cosets of the subgroup $H = \langle (0, 1) \rangle \subseteq \mathbb{Z}_2 \times \mathbb{Z}_2$.

3. Suppose $g^n = e$ for some element g of a group G , and positive integer n .
Prove that the order of g divides n . (Consider using the division algorithm.)